

Goals:

- Definition of basis of a topology;
- characterizations of bases;
- examples of bases;
- a topology can have several bases, while a basis generates a unique topology;
- comparing topologies via their bases.

Review:

- Let $\mathcal{A}$ be a collection of sets. A set Z can be written as a union of elements of $\mathcal{A}$ iff for every $x \in Z$ there exists $A \in \mathcal{A}$ such that $x \in A \subseteq Z$.

proof

$\Rightarrow$ Suppose $Z = \bigcup_{B \in \mathcal{B}} B$ where $\mathcal{B} \subseteq \mathcal{A}$. Let $x \in Z$. Then $x \in \bigcup_{B \in \mathcal{B}} B$

so $\exists B \in \mathcal{B} \subseteq \mathcal{A}$ s.t. $x \in B$. Clearly $B \subseteq \bigcup_{B \in \mathcal{B}} B = Z$.

$\Leftarrow$ Suppose that $\forall x \in Z \exists A_x \in \mathcal{A}$ s.t. $x \in A_x \subseteq Z$.

Then $Z = \bigcup_{x \in Z} A_x$.

Indeed, if $y \in Z$, then $\exists A_y \in \mathcal{A}$ s.t. $y \in A_y \subseteq \bigcup_{x \in Z} A_x$.

Conversely, if $y \in \bigcup_{x \in Z} A_x$, then $y \in A_x$ for some $x \in Z$

and since $A_x \subseteq Z$ we get $y \in Z$.

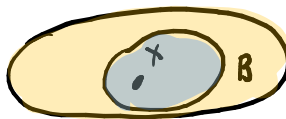
Definition 1. Let $(X, \mathcal{T})$ be a topological space. A **basis** for $\mathcal{T}$ is a collection of open sets $\mathcal{B} \subseteq \mathcal{T}$ such that every open set in $\mathcal{T}$ can be written as a union of elements of $\mathcal{B}$.

Note: Let $(X, \mathcal{T})$ be a topological space. A collection $\mathcal{B}$ of subsets of X is a basis for $\mathcal{T}$ if and only if



$$\forall A \in \mathcal{T}, \forall x \in A, \exists B \in \mathcal{B} \text{ s.t. } x \in B \subseteq A.$$

A



Example 1

- If $(X, \mathcal{T})$ is a topological space, then $\mathcal{T}$ is a basis for $\mathcal{T}$
- In a metric space, the collection of all open balls is basis for the topology induced by the metric

proof

Let X be a metric space and $\mathcal{T}$ be the topology induced by the metric. Consider $\mathcal{B} = \{B_r(x) \mid x \in X, r > 0\}$. Let $A \in \mathcal{T}$ and $x \in A$. Then $\exists r > 0$ s.t. $B_r(x) \subseteq A$, so taking $B = B_r(x) \in \mathcal{B}$ we have $x \in B \subseteq A$. This shows that $\mathcal{B}$ is a basis for $\mathcal{T}$.

- If X is a nonempty set, the collection of all singletons is a basis for the discrete topology on X .

proof

Let $\mathcal{T}$ be the discrete topology on X and $\mathcal{B} = \{\{x\} \mid x \in X\}$. Clearly, for all $A \in \mathcal{T}$ we have $A = \bigcup_{x \in A} \{x\}$, which shows that $\mathcal{B}$ is a basis for $\mathcal{T}$.

Proposition 1. Let X be a nonempty set and $\mathcal{B}$ be a collection of subsets of X satisfying the following conditions

$$1. \bigcup_{B \in \mathcal{B}} B = X,$$

2. for all $A, B \in \mathcal{B}$, the intersection $A \cap B$ can be written as a union of elements of $\mathcal{B}$.

Then, there exists a unique topology $\mathcal{T}$ on X for which $\mathcal{B}$ is a basis.

Note that, as usual, condition 2 can be replaced with the equivalent condition

2' for all $A, B \in \mathcal{B}$, for all $x \in A \cap B$, there exists $C \in \mathcal{B}$ s.t. $x \in C \subseteq A \cap B$.

proof Define $\mathcal{T}$ to be the collection of all sets that can be written as unions of elements of $\mathcal{B}$. We will show $\mathcal{T}$ is a topology.

• $X = \bigcup_{B \in \mathcal{B}} B$ by 1. and $\emptyset$ can be written as a union of the empty family of sets.

• Let $A_1, A_2 \in \mathcal{T}$. Then $A_1 = \bigcup_{i \in I} B_i$ and $A_2 = \bigcup_{j \in J} C_j$

for some families of sets $\{B_i\}_{i \in I}, \{C_j\}_{j \in J} \subseteq \mathcal{B}$.

We have $A_1 \cap A_2 = \bigcup_{i \in I} B_i \cap \bigcup_{j \in J} C_j = \bigcup_{\substack{i \in I \\ j \in J}} (B_i \cap C_j)$.

Condition 2 implies that $\forall i \in I, j \in J$, $B_i \cap C_j$ can be written as a union of elements of $\mathcal{B}$, thus $A_1 \cap A_2 \in \mathcal{T}$.

• Let $\{A_i\}_{i \in I} \subseteq \mathcal{T}$. Since A_i is a union of elements of $\mathcal{B}$ for all $i \in I$, so is $\bigcup_{i \in I} A_i$, thus $\bigcup_{i \in I} A_i \in \mathcal{T}$.

We conclude that $\mathcal{T}$ is a topology on X and $\mathcal{B}$ is a basis for $\mathcal{T}$ (by definition). Uniqueness of $\mathcal{T}$ also follows by definition of basis.

Corollary 1. Let X be a nonempty set and $\mathcal{B}$ be a collection of subsets of X satisfying the following conditions

$$1. \bigcup_{B \in \mathcal{B}} B = X,$$

2. for all $A, B \in \mathcal{B}$, we have $A \cap B \in \mathcal{B}$ or $A \cap B = \emptyset$.

Then, there exists a unique topology $\mathcal{T}$ on X for which $\mathcal{B}$ is a basis.

Proposition 2. Let X be a nonempty set and let $\mathcal{B}$ and $\mathcal{B}'$ be bases for the topologies $\mathcal{T}$ and $\mathcal{T}'$, respectively. We have that

$$\mathcal{T} \subseteq \mathcal{T}' \Leftrightarrow \text{for all } x \in X, \text{ for all } B \in \mathcal{B} \text{ that contain } x, \text{ there exists } B' \in \mathcal{B}' \text{ s.t. } x \in B' \subseteq B. \textcircled{c}$$

$\Leftrightarrow$ every element of $\mathcal{B}$ can be written as a union of elements of $\mathcal{B}'$.

proof

$\Rightarrow$ Assume $\mathcal{T} \subseteq \mathcal{T}'$. Let $x \in X$ and $B \in \mathcal{B}$ s.t. $x \in B$.

Since $\mathcal{B} \subseteq \mathcal{T} \subseteq \mathcal{T}'$, we have that $B \in \mathcal{T}'$.

Since $\mathcal{B}'$ is a basis for $\mathcal{T}'$, $\exists B' \in \mathcal{B}'$ s.t. $x \in B' \subseteq B$.

(we applied $\star$).

$\Leftarrow$ Assume $\textcircled{c}$. Let $A \in \mathcal{T}$. Since $\mathcal{B}$ is a basis for $\mathcal{T}$,

A can be written as a union of elements of $\mathcal{B}$.

Condition $\textcircled{c}$ implies that every element of $\mathcal{B}$ can be written as a union of elements of $\mathcal{B}'$.

It follows that A can be written as a union of elements of $\mathcal{B}'$, so $A \in \mathcal{T}'$. This shows $\mathcal{T} \subseteq \mathcal{T}'$.

Example 2 For $x = (x_1, x_2) \in \mathbb{R}^2$ and $r > 0$ define

$$D_r(x) = D_r(x_1, x_2) = \{(y_1, y_2) \in \mathbb{R}^2 \mid (y_1 - x_1)^2 + (y_2 - x_2)^2 < r^2\}$$

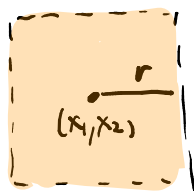
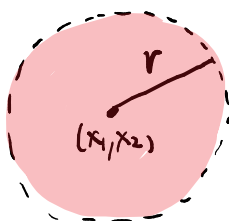
and

$$Q_r(x) = Q_r(x_1, x_2) = \{(y_1, y_2) \in \mathbb{R}^2 \mid \max\{|y_1 - x_1|, |y_2 - x_2|\} < r\}$$

Consider the following collections of subsets of the plane

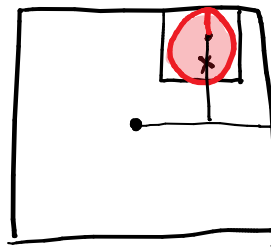
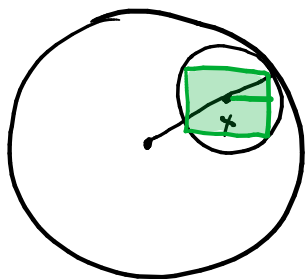
$$\mathcal{B} = \{D_r(x) \mid x \in \mathbb{R}^2, r > 0\} \quad \text{and} \quad \mathcal{R} = \{Q_r(x) \mid x \in \mathbb{R}^2, r > 0\}.$$

Then, $\mathcal{B}$ and $\mathcal{R}$ are bases for the same topology on $\mathbb{R}^2$. Which topology is it?



Note that $D_r(x)$ is the open ball in $\mathbb{R}^2$ wrt the Euclidean metric d_2 and $Q_r(x)$ is the open ball in $\mathbb{R}^2$ wrt the d_∞ metric. Thus, $\mathcal{B}$ and $\mathcal{R}$ are bases for the topologies induced by d_2 and d_∞ , respectively.

The topology induced by d_2 is the Euclidean topology. Denote this topology by τ_E and by τ the topology induced by d_∞ . Using prop 2 we can show that $\tau_E \subseteq \tau$ and $\tau \subseteq \tau_E$, so that $\tau = \tau_E$. (Rigorous proof left as exercise)



[Alternatively, we could show that d_2 and d_∞ are equivalent metrics and apply the results of hw 1, pb 5]

Example 3 On $\mathbb{R}$ consider the following collections of sets:

$$\mathcal{B}_E = \{]a, b[\mid a < b\}, \quad \mathcal{B}_L = \{[a, b[\mid a < b\}, \quad \mathcal{B}_K = \mathcal{B} \cup \{]a, b[\setminus K \mid a < b\}$$

where $K = \{\frac{1}{n} \mid n \in \mathbb{N}\}$. We have that $\mathcal{B}_E$, $\mathcal{B}_L$, and $\mathcal{B}_K$ are bases for three distinct topologies on $\mathbb{R}$ respectively called the (standard) Euclidean topology $\mathcal{T}_E$, the lower limit topology $\mathcal{T}_L$, and the K -topology $\mathcal{T}_K$. The lower limit and the K -topology are each strictly finer than the Euclidean one, and are not comparable.

We already know that $\mathcal{B}_E$ is a basis for the Euclidean topology, since $\mathcal{B}_E$ coincides with the collection of open balls in the Euclidean metric.

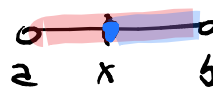
Now we check that $\mathcal{B}_L$ is the basis for a topology on $\mathbb{R}$.

Clearly $\mathbb{R} = \bigcup_{\substack{a, b \in \mathbb{R} \\ a < b}}]a, b[$. Also, the intersection

of two elements of $\mathcal{B}_L$ is either empty or an element of the form $]a, b[$ for some $a, b \in \mathbb{R}$, $a < b$. By corollary 1, $\mathcal{B}_L$ is the basis for a topology on $\mathbb{R}$.

Now we show that $\mathcal{T}_L$ is strictly finer than $\mathcal{T}_E$.

Let $x \in \mathbb{R}$, $]a, b[\in \mathcal{B}_E$ s.t. $x \in]a, b[$



Then $[x, b[\in \mathcal{B}_L$ satisfies $x \in [x, b[\subseteq]a, b[$.

By proposition 2, we get $\mathcal{T}_E \subseteq \mathcal{T}_L$.

Note that $[0, 1[\in \mathcal{B}_L \subseteq \mathcal{T}_L$, however $[0, 1[\notin \mathcal{T}_E$ (we showed that $[0, 1[$ is not open in the Euclidean topology in the past).

We conclude that $\mathcal{T}_L$ is strictly finer than $\mathcal{T}_E$.

Exercise 1

1. Prove that $\mathcal{B} = \{]p, q[\mid p < q \in \mathbb{Q}\}$ is a basis for the Euclidean topology on $\mathbb{R}$.
2. Prove that $\mathcal{B} = \{[p, q[\mid p < q \in \mathbb{Q}\}$ is not a basis for the lower limit topology on $\mathbb{R}$.

(hw)